

A positive factorization answer to Saleh's product-ideal problem

arXiv:1507.00861, Open Problem 1

Status: **full solution, likely valid; notation-sensitive but the proof covers the standard readings of $\mathfrak{B}$**

Source problem

The source is M. A. S. Saleh, *Nonlinear Operator Ideals Between Metric Spaces and Banach Spaces (PART I)*, arXiv:1507.00861. In Section 2.8, after defining products of linear and nonlinear ideals, Saleh asks:

Is it true that $\mathfrak{B} \circ \mathfrak{W}^L = \mathfrak{R}^L$?

Here $\mathfrak{R}^L$ is the nonlinear ideal of Lipschitz compact maps and $\mathfrak{W}^L$ is the nonlinear ideal of Lipschitz weakly compact maps.

Lipschitz operator BTA admits a factorization

$$BTA : X_0 \xrightarrow{\tilde{\tilde{A}}} M \xrightarrow{\tilde{\tilde{B}}} F_0,$$

where $\tilde{\tilde{B}} = B \circ \tilde{B}$ and $\tilde{\tilde{A}} = \tilde{A} \circ A$, hence the condition $(\widetilde{\mathbf{NOI}_2})$ satisfied. ■

We raise the following problem which we think is interesting.

Open Problem 1. *Is it true that $\mathfrak{B} \circ \mathfrak{R}^L = \mathfrak{R}^L$?*

2.9 Quotients of Nonlinear Ideals

Let $\mathfrak{A}$ be an operator ideal and $\mathfrak{A}^L$ be a nonlinear ideal. A Lipschitz operator $T \in Lip(X, F)$ belongs to the quotient $\mathfrak{A}^{-1} \circ \mathfrak{A}^L$ if $B \circ T \in \mathfrak{A}^L(X, F_0)$ for all $B \in \mathfrak{A}(F, F_0)$, where F_0 is an arbitrary Banach space.

The source paper uses $\mathfrak{L}$ for all bounded linear operators and does not redefine $\mathfrak{B}$ in the body. In the usual Pietsch-style notation suggested by the nearby Rosenthal/Dunford-Pettis terminology, $\mathfrak{B}$ is the ideal of completely continuous linear operators; some readers may instead parse $\mathfrak{B}$ here as the compact linear ideal. The proof below gives a compact linear factor in the nontrivial inclusion, hence proves the equality for both standard readings.

Definitions used

For a base-point preserving Lipschitz map $T : X \rightarrow F$, define its Lipschitz image by

$$\mathrm{Im}_{\mathrm{Lip}}(T) = \left\{ \frac{T(x) - T(y)}{d(x, y)} : x, y \in X, x \neq y \right\} \subset F.$$

Then $T \in \mathfrak{R}^L(X, F)$ means that $\mathrm{Im}_{\mathrm{Lip}}(T)$ is relatively norm compact, while $T \in \mathfrak{W}^L(X, F)$ means that $\mathrm{Im}_{\mathrm{Lip}}(T)$ is relatively weakly compact.

The product $\mathfrak{B} \circ \mathfrak{W}^L$ consists of all maps $T : X \rightarrow F$ admitting a factorization

$$X \xrightarrow{A} Z \xrightarrow{C} F,$$

where $A \in \mathfrak{W}^L(X, Z)$ and $C \in \mathfrak{B}(Z, F)$ is linear.

Proof intuition

A Lipschitz compact map is compact after linearization on the Lipschitz-free space. A compact linear operator admits a compact factorization through a reflexive Banach space by the compact case of the Davis-Figiel-Johnson-Pelczynski construction. Pulling this factorization back along the canonical map $\delta_X : X \rightarrow \mathcal{F}(X)$ produces the desired Lipschitz factorization: the first leg lands in a reflexive space, so its Lipschitz image is weakly compact, and the second leg is compact linear.

The reverse inclusion is the nonlinear version of the elementary fact that a compact or completely continuous linear operator maps weakly compact sets into norm relatively compact sets.

Lemma 1 (linearization of Lipschitz compact maps). *Let $T : X \rightarrow F$ be a base-point preserving Lipschitz map and let $\widehat{T} : \mathcal{F}(X) \rightarrow F$ be its Lipschitz-free linearization, $\widehat{T}\delta_X(x) = T(x)$. If $T \in \mathfrak{K}^L(X, F)$, then $\widehat{T}$ is a compact linear operator.*

Proof. The closed unit ball of $\mathcal{F}(X)$ is the closed absolutely convex hull of the molecules

$$m_{x,y} = \frac{\delta_X(x) - \delta_X(y)}{d(x,y)} \quad (x \neq y).$$

Thus $\widehat{T}(m_{x,y}) = (T(x) - T(y))/d(x,y)$, so the image of the molecule set is $\text{Im}_{\text{Lip}}(T)$.

Let $K = \overline{\text{Im}_{\text{Lip}}(T)}$, which is norm compact. The closed absolutely convex hull $\overline{\text{aco}}(K)$ is also norm compact: for each $\varepsilon > 0$, choose a finite ε -net $K_\varepsilon \subset K$; then $\text{aco}(K) \subset \text{aco}(K_\varepsilon) + \varepsilon B_F$, and the finite-dimensional compact set $\text{aco}(K_\varepsilon)$ gives total boundedness. Therefore

$$\widehat{T}(B_{\mathcal{F}(X)}) \subset \overline{\text{aco}}(K)$$

is relatively compact. Hence $\widehat{T}$ is compact. $\square$

Lemma 2 (compact DFJP factorization). *If $u : E \rightarrow F$ is a compact linear operator, then there are a reflexive Banach space Z , a bounded linear operator $S : E \rightarrow Z$, and a compact linear operator $C : Z \rightarrow F$ such that $u = CS$.*

Proof. This is the compact form of the Davis-Figiel-Johnson-Pelczynski factorization theorem. Applying the DFJP interpolation construction to the compact absolutely convex set $\overline{\text{aco}} u(B_E)$ gives a reflexive interpolation space Z and a compact inclusion $C : Z \rightarrow F$ whose image contains $u(B_E)$. The induced map $S : E \rightarrow Z$ satisfies $u = CS$. See Davis, Figiel, Johnson and Pelczynski, *Factoring weakly compact operators*, J. Funct. Anal. 17 (1974), 311–327; in the compact case the canonical inclusion produced by the construction is compact. $\square$

Theorem 1. *With $\mathfrak{B}$ interpreted as the ideal of completely continuous linear operators, Saleh’s Open Problem 1 has a positive answer:*

$$\mathfrak{B} \circ \mathfrak{W}^L = \mathfrak{K}^L.$$

The same equality holds if $\mathfrak{B}$ is read as the compact linear ideal.

Proof. First suppose $T \in \mathfrak{K}^L(X, F)$. By Lemma 1, its linearization $\widehat{T} : \mathcal{F}(X) \rightarrow F$ is compact. Lemma 2 gives a factorization

$$\widehat{T} = CS, \quad S : \mathcal{F}(X) \rightarrow Z, \quad C : Z \rightarrow F,$$

where Z is reflexive and C is compact linear. Define

$$A = S\delta_X : X \rightarrow Z.$$

Then $T = CA$. Moreover, for $x \neq y$,

$$\frac{A(x) - A(y)}{d(x, y)} = S \left(\frac{\delta_X(x) - \delta_X(y)}{d(x, y)} \right) \in S(B_{\mathcal{F}(X)}).$$

The set $S(B_{\mathcal{F}(X)})$ is bounded in the reflexive space Z , hence relatively weakly compact. Therefore $A \in \mathfrak{W}^L(X, Z)$. Since C is compact, it belongs to the compact ideal and also to the completely continuous ideal. Thus

$$T \in \mathfrak{B} \circ \mathfrak{W}^L.$$

This proves $\mathfrak{R}^L \subseteq \mathfrak{B} \circ \mathfrak{W}^L$.

Conversely, suppose $T = CA$, with $A \in \mathfrak{W}^L(X, Z)$ and $C \in \mathfrak{B}(Z, F)$. If $\mathfrak{B}$ is the compact linear ideal, then C maps bounded sets to norm relatively compact sets, so $C(\text{Im}_{\text{Lip}}(A)) = \text{Im}_{\text{Lip}}(T)$ is relatively compact.

If $\mathfrak{B}$ is the completely continuous ideal, let W be the weak closure of $\text{Im}_{\text{Lip}}(A)$. This is weakly compact. Every sequence in $C(W)$ has a preimage sequence in W ; by Eberlein-Smulian, it has a weakly convergent subsequence, and complete continuity of C sends that subsequence to a norm convergent subsequence. Hence $C(W)$ is norm relatively compact, again by sequential compactness in metric spaces. Since $\text{Im}_{\text{Lip}}(T) \subset C(W)$, $T \in \mathfrak{R}^L(X, F)$.

Thus $\mathfrak{B} \circ \mathfrak{W}^L \subseteq \mathfrak{R}^L$, and the equality follows. $\square$

Verification and limitations

The argument is purely structural. The only external tool is the compact DFJP factorization lemma, used exactly once to obtain a reflexive intermediate space and a compact final linear map. The proof does not require X to be a Banach space; it works for arbitrary pointed metric X , matching the scope of Saleh's $\mathfrak{A}^L$ -type nonlinear ideals.

The source notation $\mathfrak{B}$ is not redefined near Open Problem 1. If a reviewer confirms that Saleh intended $\mathfrak{B}$ as the completely continuous ideal, the theorem above answers the question exactly. If Saleh intended the compact linear ideal, the same proof answers that reading as well. The proof would not apply to the ideal of all bounded operators, but the paper denotes that class by $\mathfrak{L}$, not by $\mathfrak{B}$.

Novelty and literature check

Before promotion, the local run indexes were searched for 1507.00861, the title, and the terms “Lipschitz operator ideals”, “Lipschitz compact”, and “Lipschitz weakly compact”; no prior packet for this exact problem was found. Web searches on 2026-06-21 for exact and close phrases including “ $\mathfrak{B} \circ \mathfrak{W}^L = \mathfrak{R}^L$ ”, “Nonlinear Operator Ideals Between Metric Spaces and Banach Spaces” with the displayed formula, and “Lipschitz compact Lipschitz weakly compact factorization” did not reveal a direct published answer. A later parsed manuscript, arXiv:2512.06317, contains related factorization statements for dual Lipschitz p -compact classes, but it does not explicitly identify or answer Saleh's product-ideal problem.

Human-review recommendation

Review as a likely full solution. The main points to verify are:

1. Confirm Saleh's intended meaning of $\mathfrak{B}$ from Pietsch notation.
2. Check the compact version of DFJP used in Lemma 2.
3. Check that the Lipschitz-free unit-ball/molecule argument in Lemma 1 matches the exact base-point convention in Saleh's paper.

References

- M. A. S. Saleh, *Nonlinear Operator Ideals Between Metric Spaces and Banach Spaces (PART I)*, arXiv:1507.00861.
- A. Jimenez-Vargas, J. M. Sepulcre, and M. Villegas-Vallecillos, *Lipschitz compact operators*, J. Math. Anal. Appl. 415 (2014), 889–901.
- W. J. Davis, T. Figiel, W. B. Johnson, and A. Pelczynski, *Factoring weakly compact operators*, J. Funct. Anal. 17 (1974), 311–327.
- G. Godefroy and N. J. Kalton, *Lipschitz-free Banach spaces*, Studia Math. 159 (2003), 121–141.